IEEE PCB Design Workshop: (Week 05) Design Rule Checks


Altium Designer

Design Rule Verification Report

Date:	10/1/2025
Time:	11:38:27
Elapsed Time:	00:00:00
Filename:	D:\IEEE-PCB-Workshop\Power Supply Projects\Power_Supply\USB-C_Power_Supply_PCB.PcbDoc

Warnings:	0
Rule Violations:	0

Summary

Warnings	Count
Total	**0**

Rule Violations	Count
Clearance Constraint (Gap=0.254mm) (All),(All)	0
Short-Circuit Constraint (Allowed=No) (All),(All)	0

Hosted By: Adrian Sucahyo and IEEE at the University of Utah
Adapted From: IEEE x FSAE Workshop SP25 with Nick Howard and Adrian Sucahyo

Workshop Outline

Tentative Schedule:

- Sept. 3 – Introduction to Schematics
- Sept. 10 – Schematics and Components
- Sept. 17 – Introduction to PCB Layout
- Sept. 24 – Footprints and PCB Layout
- Oct. 1 – Open Work Session
- ** FALL BREAK **
- Oct. 22 – Soldering Week 1
- Oct. 29 – Soldering Week 2
- Nov. 5 – Soldering Week 3
- Nov. 12 – Final Notes and Next Steps

Announcements

- ASUU Budget
  - We have been approved!
  - We will NOT have a fee to get your board manufactured for the soldering portion of the workshop

- Alternative Projects
  - We will be able to get alternative project boards manufactured if submitted by the deadline.
  - Limited to the 10 cm x 10 cm dimensions outlined by JLCPCB.
  - Talk or email me if you have any questions!

Announcements

- Board Submission Deadline!
  - October 3rd, 11:59 PM
  - Submit Gerber files to get them manufactured with the reference design

DEPARTMENT OF
Electrical & Computer Engineering
COLLEGE OF ENGINEERING | THE UNIVERSITY OF UTAH

Want more experience?

- Consider joining the FSAE tractive team!
  - The Tractive Team is currently looking for students to assist with designing and assembling the electrical system for an electric formula-style race car!
  - No experience required!

U of U FSAE Discord Link

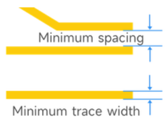
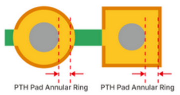
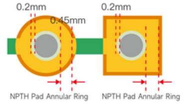
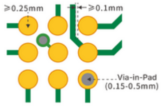
Join the IEEE Discord

- If you haven’t already, please join the IEEE Discord server for additional information and updates regarding this workshop

U of U IEEE Discord Link

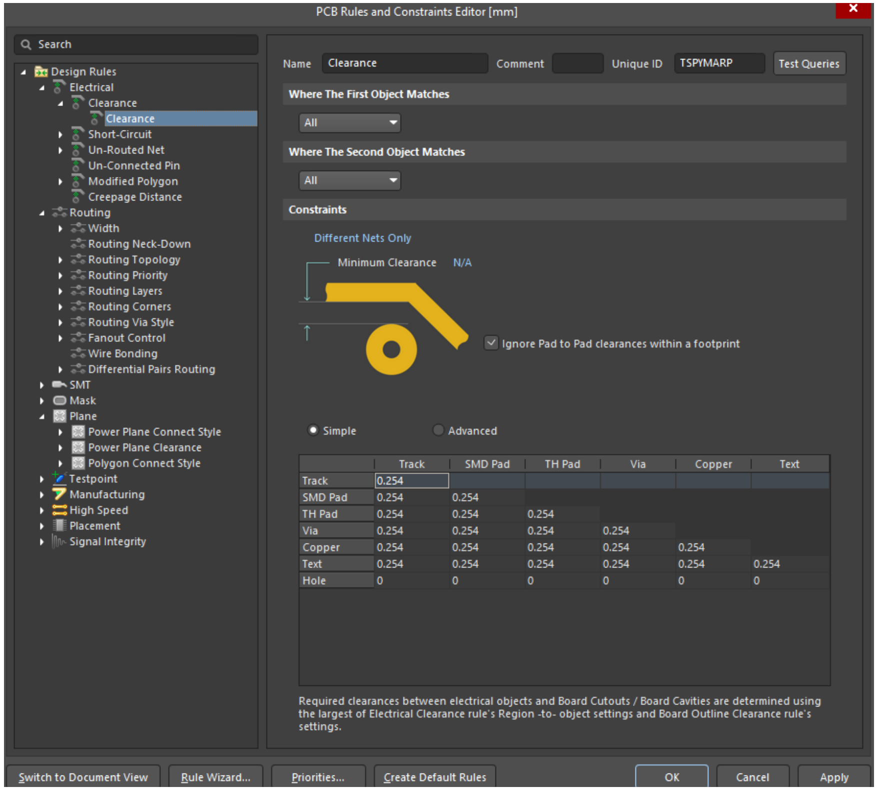
Design Rules

- Design Rules are what the manufacturer is capable of
  - All components for a project must be derived from a library
  - Projects may reference multiple different component libraries
  - Relatively consistent across platforms
- Typically, there are symbol and footprint libraries
  - Tightly coupled together

Features	Capability	Description	Patterns
Min. track width and spacing (1 oz)	0.10 / 0.10 mm (4 / 4 mil)	1- and 2-layer: 0.10 / 0.10 mm (4 / 4 mil) Multilayer: 0.09 / 0.09 mm (3.5 / 3.5 mil). 3 mil is acceptable in BGA fan-outs.	
Min. track width and spacing (2 oz)	0.16 / 0.16 mm (6.5 / 6.5 mil)	2-layer: 0.16 / 0.16 mm (6.5 / 6.5 mil) Multilayer: 0.16 / 0.20 mm (6.5 / 8 mil)	
Min. track width and spacing (2.5 oz)	2 layer: 0.2/0.2mm(8mil/8mil)		
Min. track width and spacing (3.5 oz)	2 layer: 0.25/0.25mm(10mil/10mil)		
Min. track width and spacing (4.5 oz)	2 layer: 0.3/0.3mm(12mil/12mil)		
Track width tolerance	±20%	e.g. For a 0.1 mm track, the finished track width ranges from 0.08 and 0.12 mm.	
PTH annular ring	≧0.20mm	2-layer: 1 oz: Recommended 0.25 mm or above; absolute minimum 0.18 mm 2 oz: 0.254 mm or above Multilayer: 1 oz: Recommended 0.20 mm or above; absolute minimum 0.15 mm 2 oz: 0.254 mm or above	
NPTH pad annular ring	≧0.45mm	Recommended 0.45 mm or more. This is to allow a 0.2 mm ring of copper to be removed around the hole for the sealing film to attach. Pad sizes smaller than the recommended value can result in the annular ring being very thin or completely missing.	
BGA	0.25mm	① BGA pad diameter ≥ 0.25 mm ② BGA pad to trace clearance ≥ 0.1 mm (min. 0.09 mm for multilayer boards) ③ Vias can be placed within BGA pads using filled and plated-over vias	

Design Rule Check

- Altium has a built-in design rule checker
  - Can be manually configured per project
  - Import an existing rule set to use in each project.
- Run the DRC before exporting Gerbers
  - Prevents erroneous or unmanufacturable boards from being sent



Questions?

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Forms

PCB Submission Form

https://forms.gle/PixYUWDKjhwrTAVo6

Workshop Feedback

https://forms.gle/STEPmcGkChKPyNLA6